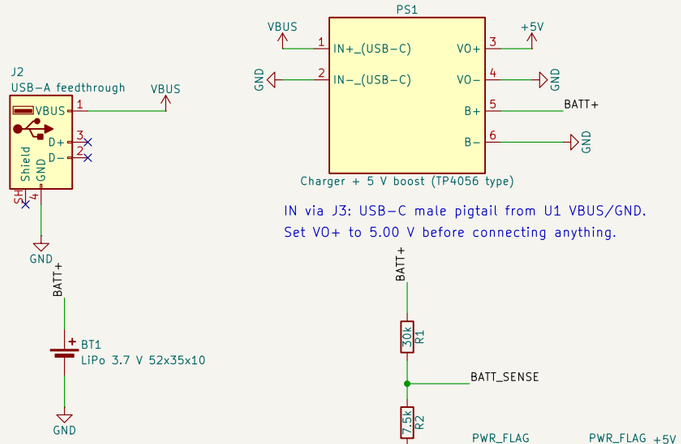
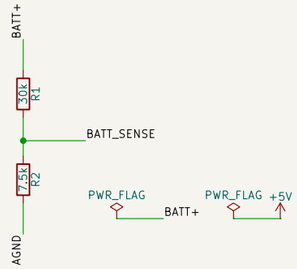
POWER, CHARGING, BATTERY SENSE

J2 = the rear USB-A port. D+/D- go to U1's micro-USB jack
through a short micro-USB-to-A cable (not drawn).



IN via J3: USB-C male pigtail from U1 VBUS/GND.
Set VO+ to 5.00 V before connecting anything.



U2 sensor module (R1/R2): BATT+ / 5.
GND screw empty; S->GP26, - ->AGND.

BUTTONS (JZT-08 LIT BUTTONS, 3-PIN XH)

SW1 plug: pin 1 = LED-, pin 2 = switch -> BTN_LEFT, pin 3 = LED+/COM -> BTN_COM



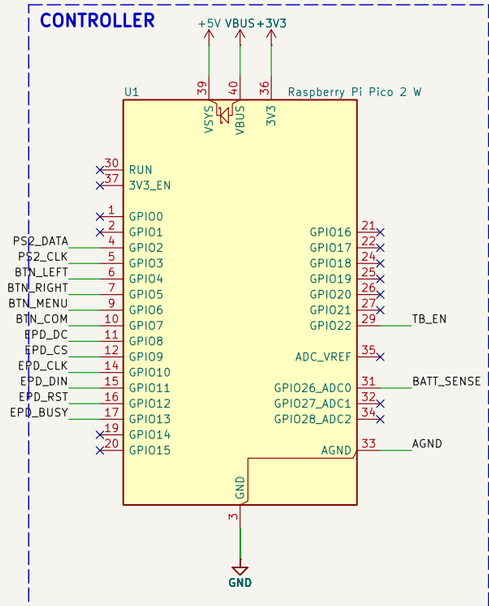
SW2 plug: pin 1 = LED-, pin 2 = switch -> BTN_RIGHT, pin 3 = LED+/COM -> BTN_COM



SW3 plug: pin 1 = LED-, pin 2 = switch -> BTN_MENU, pin 3 = LED+/COM -> BTN_COM

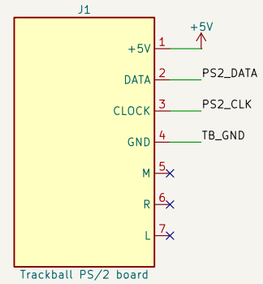
BTN_COM (GP7) is driven HIGH to light the LEDs and pulsed LOW for 20 us to read
the switches on GP4/5/6 (internal pull-ups, active low).

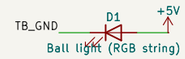
CONTROLLER

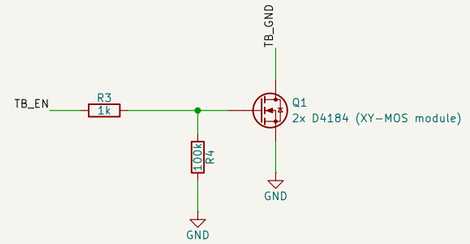


U1: Pico 2 W, mounted pins-up.

TRACKBALL + SLEEP SWITCH

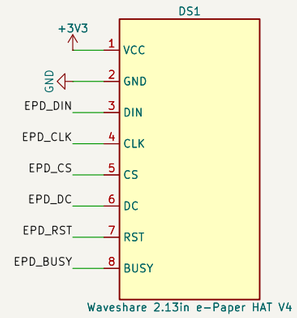






Q1 = MOSFET module: OUT- (drain) is the trackball's switched ground TB_GND.
TB_EN high = trackball on; low in sleep. R3/R4 are on the module; VIN+/OUT+ just pass +5V.

E-PAPER DISPLAY



3.3 V only. 8-pin JST lead.

GT TRACKBALL — SCHEMATIC. Net labels with the same name are connected.

Wiring harness (wire colours, lengths, plugs): harness/gt-trackball.svg

Licence: CC BY-SA 4.0
Wireless 3in arcade trackball, Pico 2 W
github.com/growthlayer/gt-trackball

Sheet: /
File: gt-trackball.kicad_sch

Title: GT Trackball

Size: A3	Date: 2026-10-12	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1